ELECTRONIC SPEED CONTROLLER

ARK 4IN1 ESC CONS

Connectorized NDAA-compliant 4-in-1 ESC running ARK32 firmware

EXPOSED INTERFACES DSHOT PWM TELEM SWD

The USA-built ARK 4IN1 ESC CONS is the connectorized variant of the ARK 4IN1 ESC. Four independent STM32F051 MCUs drive a 3-phase power stage under ARK32 firmware (supports AM32). Built-in motor and battery connectors eliminate soldering.

Board capability remains 50 A continuous / 75 A burst per motor; MR30 motor connectors limit practical continuous current to 30 A / 40 A burst. Includes 10 AWG power cables with M2.5 ring terminals.

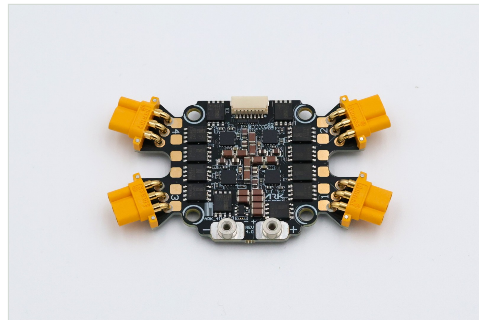
01 KEY FEATURES

- Board: 50 A continuous / 75 A burst per motor
- Connector limit: 30 A continuous / 40 A burst per motor
- 3–8S LiPo (6 V min, 65 V abs max)
- 4× independent STM32F051 channels
- Onboard current sensor (100 A/V)
- ARK32 firmware (supports AM32)
- No soldering required — MR30 motors + battery cables included

02 SPECIFICATIONS

ELECTRICAL	
Input voltage	6 V – 65 V (3–8S)
Board continuous	50 A per motor
Board burst	75 A per motor
Connector continuous	30 A per motor (MR30)
Connector burst	40 A per motor
Current sense	100 A/V
Firmware	ARK32 (supports AM32)
Operating temperature	−40 °C to +85 °C

MECHANICAL	
Dimensions	77.0 × 42.0 × 9.4 mm
Weight	24.0 g
Mounting	30.5 mm pattern



AT A GLANCE	
Voltage	3–8S (6–65 V)
Board current	50 A / 75 A
Connector limit	30 A / 40 A
Firmware	ARK32 (AM32)
Sense	100 A/V
Dimensions	77 × 42 × 9.4 mm
Weight	24.0 g
Hardware rev	Rev 4.2

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- DIU Blue Framework listed

ORDERING & RESOURCES

STORE	arkelectron.com/product/ark-4in1-esc-cons/
DOCS	docs.arkelectron.com/.../ark-4in1-esc
CAD	github.com/ARK-Electronics/ark-hardware

03 FLIGHT CONTROLLER CONNECTOR

PIN	SIGNAL	NOTES
1	VBAT	Battery voltage
2	CURR	Current sense output
3	TELEM	Serial telemetry
4	MOTOR 1	DShot / PWM
5	MOTOR 2	DShot / PWM
6	MOTOR 3	DShot / PWM
7	MOTOR 4	DShot / PWM
8	GND	

Connector: 8-pin JST-SH.

04 DEBUG CONNECTOR

PIN	SIGNAL	NOTES
1	3.3 V	
2–9	SWDIO / SWDCLK 1–4	Per-motor SWD
10	GND	

Connector: 10-pin JST-SH.

05 NOTES

- Before first use, configure KV and pole count for each motor.
- DShot is the recommended input protocol. PWM requires calibration.
- Motor connections: 4× MR30. Battery: M2.5 threaded standoffs + included 150 mm 10 AWG cables with ring terminals.

06 VARIANT

PRODUCT	KEY DIFFERENCE
ARK 4IN1 ESC	Solder-pad motors/battery; smaller 43 × 40.5 × 7.6 mm, 14.5 g; full 50 A / 75 A board rating

ARK-DS-ARK-4IN1-ESC-CONS | Issue 2026-08 | Compiled from the ARK 4IN1 ESC Production schematic Rev 4.2, the Production bill of materials, manufacturing documentation, and ARK Electronics published documentation. Board current ratings are thermal limits; connector ratings are additional limits of the MR30 connectors. Specifications are subject to change without notice.